

SOFIA BOTTI

Curriculum Vitae

PERSONAL INFORMATION

Date of birth	September 1995
Nationality	Italy
Email	sofia.botti@polimi.it
Website	bottiso.github.io
LinkedIn	sofia-botti-594570280
ORCID	0000-0002-7271-6943

CURRENT POSITION

Jan 2026 – present	Contract Researcher Politecnico di Milano, Department of Mathematics – MOX <i>FIS grant SYNERGIZE</i> , grant no. FIS-2023-02228 Milan, Italy
-----------------------	---

EMPLOYMENT HISTORY

Aug 2025 – Jul 2026	Lecturer, Module “Algorithmics” UniDistance Suisse, Faculty of Mathematics and Computer Science Brig, Switzerland
------------------------	--

Nov 2023 – Oct 2025	Postdoctoral Researcher Università della Svizzera Italiana <i>SNSF Postdoctoral Fellowship</i> , grant no. 217025 Lugano, Switzerland
------------------------	---

Nov 2022 – Oct 2023	Postdoctoral Assistant University of Pavia & Università della Svizzera Italiana <i>H2020-EuroHPC Microcard project</i> , grant no. 955495 Pavia, Italy & Lugano, Switzerland
------------------------	--

EDUCATION

Nov 2022	PhD in Computational Mathematics and Decision Sciences Joint PhD Program UNIPV-USI, 2019–2022 Thesis: <i>Mathematical modeling of Human Induced Pluripotent Stem Cell-Derived Cardiomyocytes (hiPSC-CMs): from ionic currents to 3D ventricle models</i> Supervisor: Prof. Luca F. Pavarino
----------	---

Jul 2019	Master in Mathematics University of Pavia, Department of Mathematics, 2017–2019 Thesis: <i>Mathematical and numerical modeling of Human Induced Pluripotent Stem Cell-Derived Cardiomyocytes (hiPSC-CMs)</i> Supervisor: Prof. Luca F. Pavarino Mark: 110/110 <i>cum laude</i>
----------	---

Jul 2017	Bachelor in Mathematics University of Pavia, Department of Mathematics, 2014–2017 Thesis: <i>Dynamic Systems: Topological Equivalence and Examples of Bifurcations</i> Supervisor: Prof. Giuseppe Savaré Mark: 103/110
Jun 2014	Secondary School Certificate Scientific Lyceum Giovanni Gandini, Lodi, 2009–2014 Mark: 100/100 <i>cum laude</i>

FUNDING AND GRANTS

Nov 2023 – Oct 2025	Swiss National Science Foundation (SNSF), Postdoctoral Fellowship Cell-by-cell and bidomain models for cardiac derived stem cell tissue: innovative numerical and deep learning tools for regenerative medicine PI: Sofia Botti · Grant no. 217025 · approx. 211 450 CHF
------------------------	---

SCIENTIFIC VISITING PERIODS

May 2026	Università degli Studi di Trento <i>Dipartimento di Matematica</i> , Trento, Italy Host: Prof. Simone Pezzuto
Nov 2025	King Abdullah University of Science and Technology (KAUST) <i>Computer, Electrical and Mathematical Science and Engineering Division</i> Thuwal, Saudi Arabia Host: Prof. Rolf Krause
Nov 2024	Tampere University <i>Faculty of Medicine and Health Technology</i> , Tampere, Finland Host: Prof. Jari Hyttinen
Apr 2022 – Jun 2022	Ente Ospedaliero Cantonale (EOC) – Cardiocentro Ticino Bellinzona – Lugano, Switzerland Host: Prof. Lucio Barile & Dr. Claudia Altomare
Apr 2021 – Oct 2021	Università della Svizzera Italiana (USI) <i>Euler Institute, Faculty of Informatics</i> , Lugano, Switzerland Host: Prof. Rolf Krause

SUPERVISION OF JUNIOR RESEARCHERS

Co-supervised Ph.D. students

Sep 2024 – present	Tonali Sofia Computational modeling of hiPSC-CMs MEAs Ph.D. in Computational Mathematics, Learning, and Data Science Università degli Studi di Pavia
-----------------------	--

Co-supervised Master theses

Oct 2025 – present	Rorri Foteini Learning Inverse mappings in electrophysiology at the micron scale via the EMI model MSc in Artificial Intelligence, Università della Svizzera Italiana
-----------------------	---

Oct 2023 – **Tonali Sofia**
Apr 2024 Mathematical modeling of Multielectrode Arrays with Pluripotent Stem Cell-Derived Cardiomyocytes
MSc in Mathematics, Università degli Studi di Pavia

TEACHING ACTIVITIES

Lecturer

Feb 2026 – **Algorithmics**
Jul 2026 BSc course, spring semester
Department of Mathematics, FernUni-Unidistance, Brig, Switzerland
Aug 2025 – **Algorithmics**
Jan 2026 BSc course, fall semester
Department of Mathematics, FernUni-Unidistance, Brig, Switzerland

Lecturer in Ph.D. Courses

Jun 2025 **Modeling the electrophysiology of the cardiac tissue**
Intensive Summer School on Computational Cardiology: theory, applications and new trends
Università degli Studi di Milano, Italy

Substitute Lecturer

Oct 2022 – **Sistemi Dinamici**
Jan 2023 MSc course, Department of Industrial Engineering and Department of Mathematics
Università degli Studi di Pavia, Italy

Teaching Assistant

Oct 2022 – **Analisi 1**
Jan 2023 BSc course, Department of Industrial Engineering and Information
Università degli Studi di Pavia, Italy
Oct 2021 – **Analisi 1**
Jan 2022 BSc course, Department of Industrial Engineering and Information & Department of Civil Engineering and Architecture
Università degli Studi di Pavia, Italy
Dec 2020 – **Analisi 1 – Didactic Seminars**
Feb 2021 BSc course, Department of Industrial Engineering and Information
Università degli Studi di Pavia, Italy
Nov 2020 – **Mathematics – Precourse**
Dec 2020 Faculty of Nursing, Università degli Studi di Pavia, Italy

PRIZES, AWARDS AND FELLOWSHIPS

Sep 2024 **Travel Grant**, Cardiac Physiome Workshop 2024
500 €

ORGANIZATION OF CONFERENCES AND EVENTS

Sep 2025	Organizer , Mini-symposium <i>Recent Advances in Mathematical and Computational Methods for Cardiac Biomechanics</i> ENUMATH 2025, Heidelberg, Germany
Jun 2025	Co-organizer , <i>Intensive Summer School on Computational Cardiology: theory, applications and new trends</i> Università degli Studi di Milano, Italy
May 2024	Co-organizer , ICAM Workshop <i>International Center for Advanced Computing in Medicine</i> Università della Svizzera Italiana, Switzerland

PERSONAL SKILLS

Languages	Italian (mother tongue), English (fluent), French (basic)
Operating systems	Linux, macOS, Microsoft Windows
Programming	MATLAB, C/C++, Python
Technical tools	Bash, Vim, Git, Docker
Visualization	ParaView
Documentation	L ^A T _E X/TikZ/PGFPlots/Beamer, Microsoft Office, Keynote

PUBLICATIONS

SOFIA BOTTI

PUBLICATIONS

Journal Articles

- [J7] S. Tonali, **S. Botti**, L. F. Pavarino, Second-Order IMEX Time-Stepping Methods for Efficient Bidomain Multi-Electrode Array Simulations of Cardiac Stem Cell Monolayers, *Under review*. Preprint: <https://arxiv.org/abs/2609.12650>
- [J6] **S. Botti**, M. Favino, R. Krause, C. Altomare, L. Barile, A Correlation Analysis of Field Potential and Action Potential Biomarkers in Heterogeneous hiPSC-derived Cardiac Tissues, *Under review*. Preprint: <https://doi.org/10.21203/rs.3.rs-9787691/v1>
- [J5] **S. Botti**, M. Favino, A Numerical Comparison of Multi-Electrode Array Model and Surrogate Approaches for Derived Cardiac Tissues. *Under review*
- [J4] F. Giardini, C. Palandri, **S. Botti**, C. Olianti, P. Mele, N. Ballerini, R. Piersanti, M. Giacca, P. Kohl, C. M. Zgierski-Johnston, E. Cerbai, F. Regazzoni, and L. Sacconi, High-throughput, high-fidelity reconstruction of whole mouse-heart cytoarchitecture at cellular scale. *The Journal of Physiology*. (2026) <https://doi.org/10.1113/JP290463>
- [J3] **S. Botti**, R. Krause, L. F. Pavarino. *In silico* modelling of multi-electrode arrays for enhancing cardiac drug testing on hiPSC-CM heterogeneous tissues. *The Journal of Physiology* (2025). <https://doi.org/10.1113/JP287276>
- [J2] **S. Botti**, C. Altomare, M. Paci, L. Barile, R. Krause, L. F. Pavarino, S. Severi A novel ionic model for matured and paced atrial-like hiPSC-CMs integrating IK_{ur} and IK_{Ca} currents *Computers in Biology and Medicine* (2024) 180: 108899. <https://doi.org/10.1016/j.combiomed.2024.108899> Preprint: <https://www.biorxiv.org/content/10.1101/2024.01.12.574782v1>
- [J1] **S. Botti**, M. Torre. Isogeometric simulation of a derived stem cell engineered ventricle. *Advances in Computational Science and Engineering* (2023) 1(3): 298–319. <https://doi.org/10.3934/acse.2023013>

Peer-reviewed Conference Proceedings, Book Chapters

- [C4] S. Tonali, C. Altomare, L. Barile, R. Krause, L. F. Pavarino **S. Botti**. Efficient Hybrid IMEX Schemes for Modeling Multi-Electrode Arrays in Stem Cell-Derived Cardiac Tissue. *Submitted proceeding*
- [C3] **S. Botti**, R. Krause, L. F. Pavarino (2026). Advanced Bidomain Framework for Drug Testing in Heterogeneous Cardiac hiPSC Tissues. *Trends in Mathematical and Numerical Modeling of the Cardiovascular System in Springer INdAM series*, to appear
- [C2] **S. Botti**, C. Bartolucci, R. Krause, L. F. Pavarino, S. Severi. An *in silico* study of cardiac hiPSC electronic maturation by dynamic clamp. *Functional Imaging and Modeling of the Heart (FIMH 2023)*, Lecture Notes in Computer Science, vol. 13958. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-35302-4_18

- [C1] **S. Botti** et al. Numerical simulations indicate I_{K1} dynamic clamp can unveil the phenotype of cardiomyocytes derived from induced pluripotent stem cells. *Computing in Cardiology (CinC)*, Tampere, Finland (2022), pp. 1–4.
<https://doi.org/10.22489/CinC.2022.10>

PhD thesis

- [T1] **S. Botti** (2022) Mathematical modeling of Human Induced Pluripotent Stem Cell-Derived Cardiomyocytes (hiPSC-CMs): from ionic currents to 3D ventricle models, PhD thesis,
<https://iris.unipv.it/handle/11571/1467309>

PRESENTATIONS

Plenary and invited talks

- **S. Botti**, From Ionic Dynamics to Tissue Electrophysiology in Stem Derived Cardiomyocytes. **Invited seminar** at MBM Lab seminar, Trento (Italy), 2026.
- **S. Botti**, Multiscale Modeling in Computational Cardiology: From Cells to Tissue Dynamics. **Invited speaker** at CompMath2026 Spring Workshop, Pavia (Italy), 2026.
- **S. Botti**, Multiscale Modeling in Computational Cardiology: From Cells to Tissue Dynamics. **Invited speaker** at METaL seminar cycle - University of Pavia – Pavia (Italy), 2026.
- **S. Botti**, M. Favino, Multiscale Computational Modeling of Heterogeneous hiPSC-CM Cardiac Tissues. **Invited speaker** at International Workshop on Numerical and Learning Methods for PDEs, Sanya (China), 2026.
- **S. Botti**, M. Favino, Multiscale Computational Modeling of Heterogeneous hiPSC-CM Cardiac Tissues. **Invited seminar** at KAUST (Saudi Arabia), 2025.
- **S. Botti**, Chamber-Specific Ionic Models in Regenerative Cardiology: Bridging Single-Cell Precision to Tissue-Level Heterogeneity. **Invited seminar** at MET International Seminar Series, Tampere (Finland), 2024.
- **S. Botti**, *In-silico* modeling of multi-electrode arrays to enhance cardiac drug testing on heterogeneous hiPSC-CMs tissues. **Selected oral presentation** at Cardiac Physiome, Freiburg (Germany), 2024.
- **S. Botti**, Mathematical and Numerical Modeling of cardiac stem cells and regenerative medicine, **Invited speaker** at CompMath2023 Spring Workshop, Pavia (Italy), 2023.

Conference talks

- **S. Botti**, S. Pagani, F. Regazzoni, Neural Network Emulators for Cardiac Electrophysiology Modeling in Derived Stem Cardiomyocytes. GIMC SIMAI Young , Pisa (Italy), 2026.
- **S. Botti**, M. Favino, Steklov-Poincaré domain decomposition for hiPSC-CMs multi-electrode arrays models. 29th International Conference on Domain Decomposition Methods (DD29), Milano (Italy), 2025.
- **S. Botti**, R. Krause, L.F. Pavarino, Modeling hiPSC-CM Electrophysiology on Multi-Electrode Arrays: A Coupled Bidomain and Electrode Approach. Coupled Problems, Villasimius (Italy), 2025
- **S. Botti**, A novel atrial-specific ionic model for stem cells-derived CMs. 16th World Congress on Computational Mechanics and 4th Pan American Congress on Computational Mechanics (WCCM-PANACM), Vancouver (Canada), 2024.
- **S. Botti**, A novel atrial-specific ionic model for stem cells-derived CMs. (INdAM Workshop), Rome (Italy), 2024.

- **S. Botti**, Numerical modeling of cardiac derived stem cells and isogeometric simulation of an engineered tissue. European Conference on Numerical Mathematics and Advanced Applications (ENUMATH), Lisbon (Portugal), 2023.
- **S. Botti**, M. Torre, Isogeometric solvers for cardiac reaction-diffusion equations coupled with Induced Pluripotent Stem Cell ionic models, 10th International Congress on Industrial and Applied Mathematics (ICIAM), Tokyo (Japan), 2023.
- **S. Botti**, C. Bartolucci, C. Altomare, et al., Numerical Simulations Indicate IK1 Dynamic Clamp Can Unveil the Phenotype of Cardiomyocytes Derived from Induced Pluripotent Stem Cells, Computing in Cardiology (CinC), Tampere (Finland), 2022.
- **S. Botti**, From cardiac stem cell ionic models to isogeometric simulations of 3D cardiac tissue, CompMath2022 Spring Workshop, Pavia (Italy), 2022.
- **S. Botti**, M. Torre, Isogeometric simulations of hiPSC-CM cardiac tissue, Young Researchers Workshop on Mathematical and Numerical Cardiac Modeling, Pavia (Italy), 2021.

Posters

- [P3] **S. Botti**, M. Favino, et al., A Correlation Analysis of Field Potential and Action Potential Biomarkers in Heterogeneous hiPSC-derived Cardiac Tissues, LS2 Cardiovascular Research Meeting 2026, Bern (Switzerland), 2026.
- [P2] **S. Botti**, R. Krause, et al., Modelling Atrial Fibrillation in Stem Cell-Derived CMs: *in-silico* approaches for Personalized Medicine, Gordon Research Conference (GRC), Lucca (Italy), 2025.
- [P1] **S. Botti**, C. Bartolucci, R. Krause, et al., An *in silico* Study of Cardiac hiPSC Electronic Maturation by Dynamic Clamp, Functional Imaging and Modeling of the Heart (FIMH), Lyon (France), 2023.

OTHER PRODUCTS WITH DOCUMENTED USE

- [G1] MEA_model, Open-source MATLAB framework for 2D Multi-Electrode Array (MEA) simulations of heterogeneous hiPSC-CM cardiac tissues. [Code repository](#) (Co-core developer) DOI:10.5281/zenodo.18593287
- [G1] AL_hiPSC_ionic_model, Open-source MATLAB code for the atrial-specific ionic model for human induced pluripotent stem cells. [Code repository](#) (Core developer) DOI:10.5281/zenodo.18589161